impl TopKMeasure for MultiDP

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This document proves soundness of TopKMeasure for MultiDP.

1 Hoare Triple

Precondition

Compiler-verified

- Method privacy_map: types are consistent with the PureDP and zCDP top-k accounting implementations.

Caller-verified

- Method privacy_map: d_in is non-null and non-negative, and scale is non-null and non-negative.

Postcondition

Theorem 1.1. The implementation is consistent with the associated items in the TopKMeasure trait. For any x, x' where $d_{\text{in}} \geq d_{\text{Range}}(x, x')$, privacy_map returns a PrivacyGuarantee containing a valid bound for the selected top-k mechanism.

Proof. For the exponential distribution, the implementation uses the established PureDP bound $\epsilon = kd_{\text{in}}/\text{scale}$ and embeds it as an ApproxDP profile. For the Gumbel distribution, it uses the established zCDP bound $\rho = k(d_{\text{in}}/\text{scale})^2/8$. Both bounds are composed over the k peeling rounds, and the corresponding representation is attached to the aggregate MultiDP guarantee without weakening the certified bound. Directed arithmetic rounds the final multiplication upward. □

References